

Software Design Document

1.1 Purpose

This document provides a comprehensive design of the RRI online Technical Journal— a web-based platform for managing scholarly journal activities including manuscript submission, peer review, editorial workflow, publication, and public access.

1.2 Scope

- Multi-role system (Admin, Editor, Author, Reviewer, Public)
- Full manuscript lifecycle from submission to publication
- Payment integration (SSLCommerz)
- Google analytics integration.
- Public website with dynamic content
- Analytics and audit log details.
- Plagiarism & AI detection for reviewer dashboard(System Build)

2. System Overview

2.1 High-Level Architecture

- **Frontend:** Next.js 16 (App Router) + TypeScript + Tailwind CSS
- **Backend:** Django 6.0 + Django REST Framework
- **Database:** MySQL
- **Authentication:** JWT (SimpleJWT)
- **File Storage:** Django Media + Static files
- **Payment Gateway:** SSLCommerz
- **Email Service:** Django SMTP

System Requirements Specification:

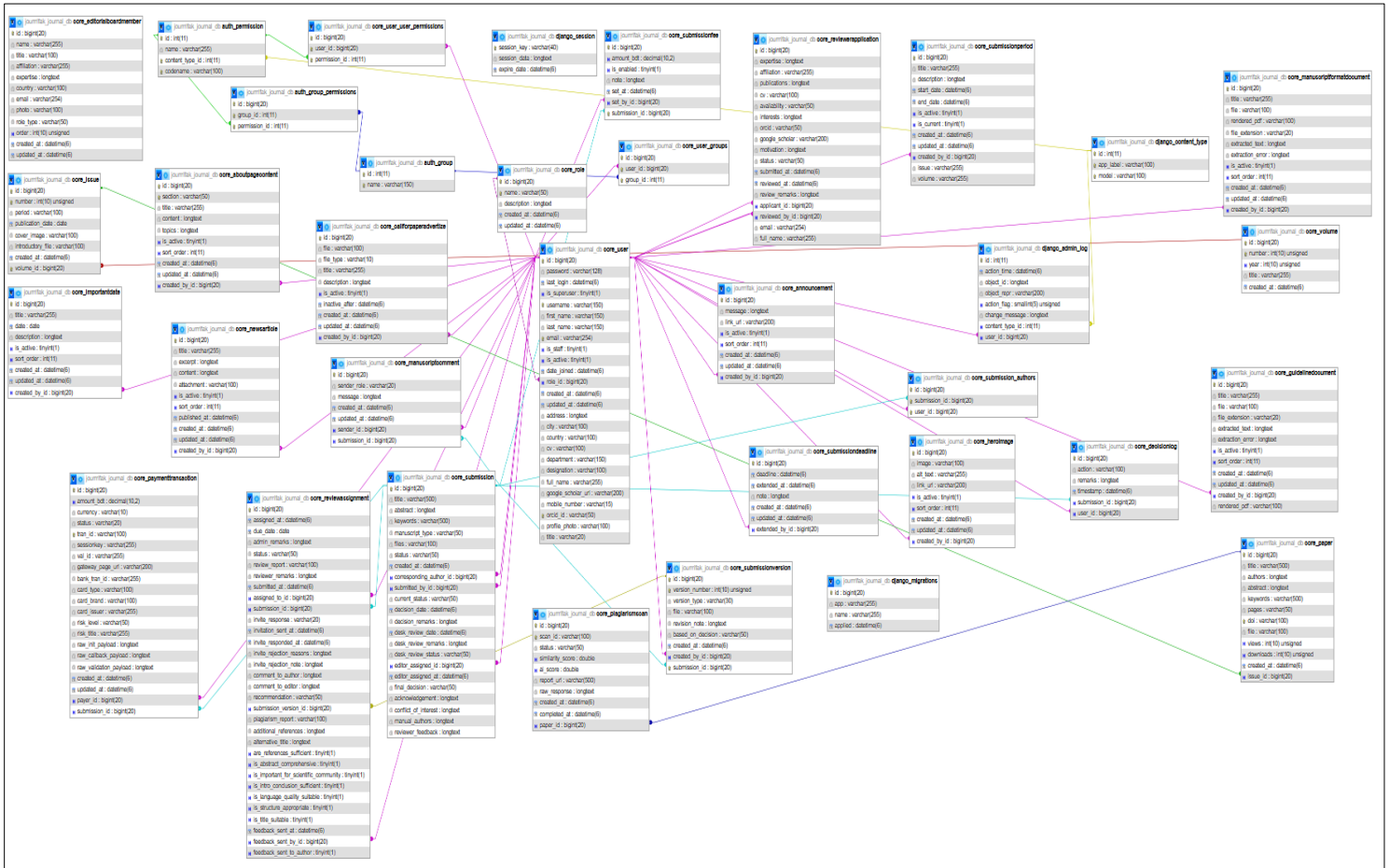
User Requirements

Role	Primary Goals	Key Requirements
Admin	Full system control & oversight	1. Manage all users & roles 2. View & control all submissions 3. Manage Call for paper advertisement. 4. Manage Top announcement bar content 5. Manage home page sliding images. 6. Manage Author Guidelines. 7. Manage About Page content. 8. Manage News Sections. 9. Manage manuscript submission deadline. 10. Manage User roles. 11. Manage editorial board members. 12. Manage authors submitted manuscripts. 13. Assign Reviewers for reviewing manuscripts. 14. Check peer review reports. 15. Analysis reviewer performance. 16. Analysis reviewer performance. 17. Able to check Google Analytics for website performance and overview. 18. Able to publish new articles under specific issues and volumes. 19. Manage APC (article processing charges). 20. Check Decision logs. 21. Update password. 22. Send call for paper invitation to authors.

Editor	Manage editorial workflow & Contents	1. Manage all users & roles 2. View & control all submissions 3. Manage Call for paper advertisement. 4. Manage Top announcement bar content 5. Manage home page sliding images. 6. Manage Author Guidelines. 7. Manage About Page content. 8. Manage News Sections. 9. Manage manuscript submission deadline. 10. Manage User roles. 11. Manage editorial board members. 12. Manage authors submitted manuscripts. 13. Assign Reviewers for reviewing manuscripts. 14. Check peer review reports. 15. Analysis reviewer performance. 16. Analysis reviewer performance. 17. Able to check Google Analytics for website performance and overview. 18. Able to publish new articles under specific issues and volumes. 19. Manage APC (article processing charges). 20. Check Decision logs. 21. Update password. 22. Send call for paper invitation to authors.
Author	Submit & track manuscripts	1. Able to make self-registration in journal system and do self-verification to get the access of user account. 2. Submit manuscript with co-authors 3. Check all submissions and get automatic mail notification for status update. 4. Pay processing fee (if applied by authority) 5. Can see Reviewers Comment (If sent by editor)

		6. Able to reset password if forgotten. 7. Able to change password and update profile information.
Reviewer	Review assigned manuscripts and give feedback	1. Able to apply become reviewer. 2. Able to access account if his application approved by journal authority. 3. Receive mail notification for application feedback. 4. Able to view the abstract in the email and decide whether to accept or reject the review invitation. In case of rejection, the reviewer must fill out a form stating the reason for rejection. 5. View assigned manuscripts on dashboard 6. Submit detailed review with checklist & recommendation, with review report and plagiarism report 7. Can comments for author. (author will can see the comments if the authority sent to author) 8. Able to update profile and password
Public User	Access published content	1. View homepage. 2. View journal website contents 3. Browse published papers, volumes & issues 4. Read guidelines & announcements 5. Search paper in archives. 6. View call for paper ad. 7. Able to make registration as Author .

Entity Relationship Diagram:



Test Server Details

Multi role user credentials:

Role	User mail	Password
admin	admin@gmail.com	12345678
Editor	habiburhemal6518@gmail.com	1234
Author	mahmud@gmail.com	1234
Reviewer	redwan.websoftbd@gmail.com	1234